# Variational Inference: KL Divergence, Jensen, and the ELBO

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## §1.1 The approximate posterior as an optimization problem

The true posterior in a latent variable model is

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)}.$$

It is usually intractable because the marginal likelihood

$$p_\theta(x) = \int p_\theta(x, z)\, dz$$

is hard to compute.

Variational inference introduces a tractable family

$$q_\phi(z \mid x)$$

and tries to make it close to the true posterior. A natural measure of closeness is

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

This is the reverse direction from what one might first guess: we place expectation under $q_\phi$, because $q_\phi$ is the distribution from which we can sample and whose density we can evaluate.

The obstacle is that this KL divergence contains $p_\theta(z \mid x)$, which itself contains the intractable $p_\theta(x)$. The central algebraic trick is to rewrite it so that $\log p_\theta(x)$ appears as a constant plus a lower bound.

## §1.2 The ELBO identity

Start from the KL divergence:

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)) = \mathbb{E}_{q_\phi(z|x)} \left[ \log \frac{q_\phi(z \mid x)}{p_\theta(z \mid x)} \right].$$

Using

$$p_\theta(z \mid x) = \frac{p_\theta(x, z)}{p_\theta(x)},$$

we get

$$\begin{aligned} D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)) &= \mathbb{E}_{q_\phi} \left[ \log q_\phi(z \mid x) - \log p_\theta(x, z) + \log p_\theta(x) \right] \\ &= \log p_\theta(x) - \mathbb{E}_{q_\phi} \left[ \log p_\theta(x, z) - \log q_\phi(z \mid x) \right]. \end{aligned}$$

Define

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)} \left[ \log p_\theta(x, z) - \log q_\phi(z \mid x) \right].$$

Then the identity becomes

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

Since KL divergence is nonnegative,

$$\mathcal{L}(\theta, \phi; x) \leq \log p_\theta(x).$$

This is why $\mathcal{L}$ is called the evidence lower bound, or ELBO.

**Remark 1.2.1 —** The ELBO is not introduced by magic. It is exactly what remains when the intractable posterior KL is separated from the log marginal likelihood.

## §1.3 Jensen's inequality derivation

The same lower bound can be derived directly from Jensen's inequality. For the equality in the next display we assume $p_\theta(x, z) = 0$ almost everywhere that $q_\phi(z \mid x) = 0$: the trial density covers the support of the unnormalized target. (A Gaussian trial density with strictly positive variance has full support and satisfies this in ordinary continuous latent models.) If $q$ misses some posterior mass, the intervening equality is replaced by an inequality, and Jensen still supplies a lower bound.

$$\begin{aligned}
\log p_\theta(x) &= \log \int p_\theta(x, z)\, dz \\
&= \log \int q_\phi(z \mid x) \frac{p_\theta(x, z)}{q_\phi(z \mid x)}\, dz \\
&= \log \mathbb{E}_{q_\phi(z|x)} \left[ \frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right].
\end{aligned}$$

Because log is concave,

$$\begin{aligned}
\log p_\theta(x) &\geq \mathbb{E}_{q_\phi(z|x)} \left[ \log \frac{p_\theta(x, z)}{q_\phi(z \mid x)} \right] \\
&= \mathbb{E}_{q_\phi(z|x)} \left[ \log p_\theta(x, z) - \log q_\phi(z \mid x) \right] \\
&= \mathcal{L}(\theta, \phi; x).
\end{aligned}$$

The two derivations emphasize different things. The KL derivation shows the gap between the bound and the true log-likelihood. The Jensen derivation shows why a lower bound exists at all.

## §1.4 Reconstruction term and regularization term

Using the latent variable factorization

$$p_\theta(x, z) = p(z) p_\theta(x \mid z),$$

the ELBO becomes

$$\begin{aligned}
\mathcal{L}(\theta, \phi; x) &= \mathbb{E}_{q_\phi(z|x)} \left[ \log p_\theta(x \mid z) + \log p(z) - \log q_\phi(z \mid x) \right] \\
&= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)).
\end{aligned}$$

This is the familiar VAE decomposition:

$$\boxed{\mathcal{L} = \text{expected log-likelihood} - \text{KL to the prior.}}$$

The first term asks whether latent samples from $q_\phi(z \mid x)$ explain $x$ well under the decoder. The second term asks whether the approximate posterior stays close to the prior.

It is common to call the first term the reconstruction term. But mathematically it is an expected log-likelihood. It becomes a squared-error reconstruction loss only under a Gaussian decoder with fixed variance. It becomes a cross-entropy-like loss under a Bernoulli decoder.

## §1.5 The ELBO gap

The identity

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x))$$

shows that there are two goals mixed together.

Increasing $\mathcal{L}$ can improve the model likelihood $\log p_\theta(x)$. But it also tries to reduce the posterior approximation gap

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

If $q_\phi$ were flexible enough to equal the true posterior, then the bound would be tight:

$$\mathcal{L}(\theta, \phi; x) = \log p_\theta(x).$$

In practice, $q_\phi$ is restricted, often to a diagonal Gaussian, so the bound is generally not exact.

## §1.6 A Gaussian ELBO Gap

We can measure how loose an ELBO is without training a neural network. Consider the continuous latent model

$$Z \sim \mathcal{N}(0, 1), \qquad X \mid Z = z \sim \mathcal{N}(z, 1),$$

and observe $x = 2$. Completing the square gives

$$\begin{aligned} -\frac{z^2}{2} - \frac{(2-z)^2}{2} &= -(z-1)^2 - 1, \\ Z \mid X = 2 &\sim \mathcal{N}(1, \tfrac{1}{2}), \\ X &\sim \mathcal{N}(0, 2). \end{aligned}$$

The exact log evidence is therefore

$$\log p(2) = -1 - \frac{1}{2}\log(4\pi).$$

Choose a variational family of Gaussians with the correct posterior variance but a movable mean:

$$q_\mu(z \mid 2) = \mathcal{N}(\mu, \tfrac{1}{2}).$$

The KL divergence between two normals of equal variance $v$ is $(m_1 - m_2)^2/(2v)$. Here $v = 1/2$, so

$$\boxed{D_{\mathrm{KL}}(q_\mu \| p(z \mid 2)) = (\mu - 1)^2.}$$

The ELBO identity now describes a complete curve, not just an inequality:

$$\boxed{\mathcal{L}(\mu; 2) = \log p(2) - (\mu - 1)^2.}$$

At $\mu = 0$ the bound is one nat below the evidence. At $\mu = 1$ it touches the evidence exactly. In this example we can tighten the ELBO by improving the *inference* distribution while holding the *generative* model fixed.

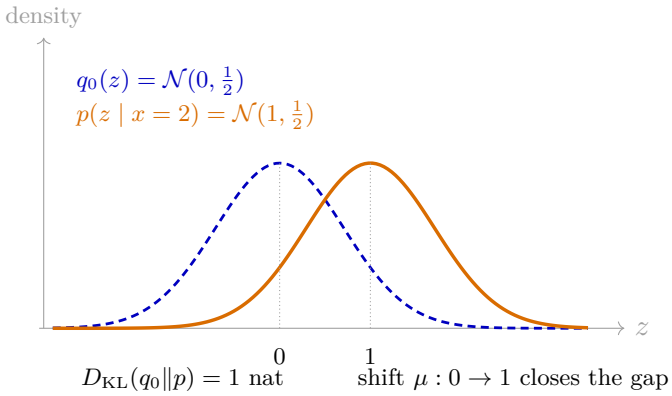



Figure 1.1: The variational gap is a mismatch of probability mass. The true posterior (solid orange) and the trial law $q_\mu = \mathcal{N}(\mu, \frac{1}{2})$ have equal variance. Moving the dashed mean from 0 to 1 makes the curves coincide and closes the one-nat ELBO gap.

This also clarifies a common inference mistake. Maximizing the ELBO over both $\theta$ and $\phi$ does *not* guarantee that every upward optimization step increases the true $\log p_\theta(x)$: the model evidence and approximation gap can change simultaneously. In a flexible posterior family, good inference makes the bound a better proxy for likelihood. In a restrictive family, model fitting and approximate inference remain entangled.[1]

## §1.7 Closed-form KL for diagonal Gaussians

The standard VAE uses

$$q_\phi(z \mid x) = \mathcal{N}(z; \mu, \mathrm{diag}(\sigma^2)), \qquad p(z) = \mathcal{N}(0, I).$$

For this pair, the KL divergence has a closed form:

$$D_{\mathrm{KL}}(q \| p) = \frac{1}{2} \sum_{j=1}^{d} \left( \mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2 \right).$$

Here is the one-dimensional calculation. Let

$$q(z) = \mathcal{N}(\mu, \sigma^2), \qquad p(z) = \mathcal{N}(0, 1).$$

[1] For the general variational derivation and its statistical interpretation, see Blei, Kucukelbir and McAuliffe, *Variational Inference: A Review for Statisticians*.

Then

$$\log q(z) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(z-\mu)^2}{2\sigma^2},$$

and

$$\log p(z) = -\frac{1}{2}\log(2\pi) - \frac{z^2}{2}.$$

Therefore

$$\begin{aligned} D_{\mathrm{KL}}(q\|p) &= \mathbb{E}_q[\log q(z) - \log p(z)] \\ &= -\frac{1}{2}\log\sigma^2 - \frac{1}{2}\mathbb{E}_q\left[\frac{(z-\mu)^2}{\sigma^2}\right] + \frac{1}{2}\mathbb{E}_q[z^2] \\ &= -\frac{1}{2}\log\sigma^2 - \frac{1}{2} + \frac{1}{2}(\mu^2+\sigma^2) \\ &= \frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log\sigma^2). \end{aligned}$$

The multivariate diagonal formula follows by summing over independent coordinates.

This is one reason the diagonal Gaussian encoder is practical: the regularization term does not need Monte Carlo estimation.

## §1.8 Negative ELBO as a loss

Training usually minimizes the negative ELBO:

$$-\mathcal{L}(\theta,\phi;x) = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] + D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z)).$$

For a Bernoulli decoder with probabilities $\pi_\theta(z)$,

$$p_\theta(x \mid z) = \prod_i \pi_i(z)^{x_i}(1-\pi_i(z))^{1-x_i},$$

so

$$-\log p_\theta(x \mid z) = -\sum_i \left[x_i \log \pi_i(z) + (1-x_i)\log(1-\pi_i(z))\right].$$

For a Gaussian decoder

$$p_\theta(x \mid z) = \mathcal{N}(x; \mu_\theta(z), \sigma^2 I),$$

we have

$$-\log p_\theta(x \mid z) = \frac{1}{2\sigma^2}\|x - \mu_\theta(z)\|^2 + \text{constant}.$$

Thus the common reconstruction losses are not arbitrary engineering choices. They are negative log-likelihoods under different decoder distributions.

## §1.9 Empirical lower bounds

The original AEVB experiments plotted the variational lower bound during training. The figure below is useful as a historical reminder: what is being optimized is not raw reconstruction error, but a lower bound $\mathcal{L}$ on marginal likelihood.

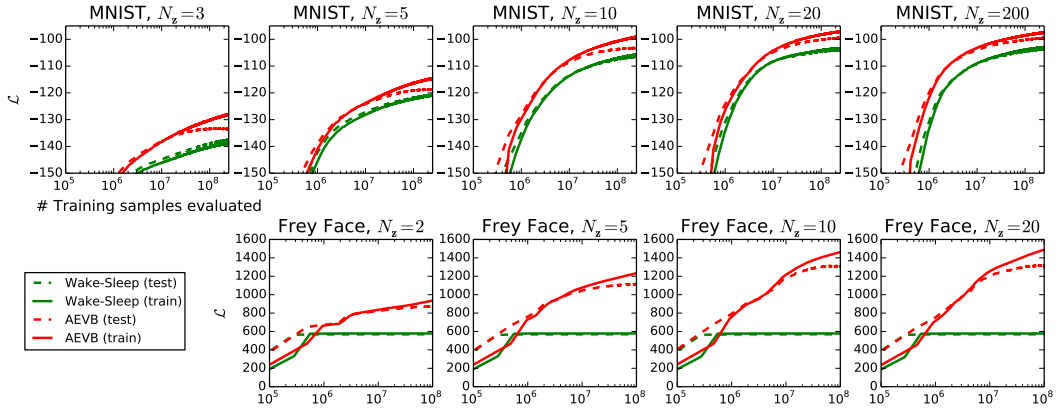



Figure 1.2: Variational lower bound curves in the AEVB experiments.

The mathematical significance of the plot comes from the identity

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z \mid x)).$$

## §1.10 Information-theoretic reading

The ELBO also resembles a rate–distortion tradeoff. The expected log-likelihood asks the latent variable to preserve enough information to explain $x$. The KL term penalizes deviations of $q_\phi(z \mid x)$ from the prior $p(z)$.

Loosely:

| term | role |
|---|---|
| $-\mathbb{E}_q[\log p_\theta(x \mid z)]$ | distortion / reconstruction cost |
| $D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z))$ | rate / information cost |

This interpretation is helpful, but it should not replace the probabilistic derivation. The ELBO is first of all a lower bound on marginal likelihood.

There is an exact identity behind the rate interpretation, and it makes the limitations of that interpretation precise. Suppose inputs are drawn from $p_{\mathrm{data}}(x)$, and write the joint law induced by the encoder as

$$q(x, z) = p_{\mathrm{data}}(x) q_\phi(z \mid x), \qquad q(z) = \int p_{\mathrm{data}}(x) q_\phi(z \mid x)\, dx.$$

The latter $q(z)$ is the *aggregated posterior*: sample a training input and then a code from its encoder distribution. Insert $\log q(z) - \log q(z)$ into the KL term averaged over inputs:

$$\begin{aligned}
&\mathbb{E}_{p_{\mathrm{data}}(x)} D_{\mathrm{KL}}(q_\phi(z \mid x) \| p(z)) \\
&= \mathbb{E}_{q(x,z)} \left[ \log \frac{q_\phi(z \mid x)}{q(z)} + \log \frac{q(z)}{p(z)} \right] \\
&= \boxed{I_q(X; Z) + D_{\mathrm{KL}}(q(z) \| p(z)).}
\end{aligned}$$

The first term is mutual information: how much the code tells us about the observed input. The second term measures how far the *overall* distribution of inferred codes is from the sampling prior. Thus the per-example KL penalty is an *upper bound* on encoded mutual information, and it also penalizes a mismatch between aggregate code usage and the prior.

A two-bit calculation distinguishes the terms. Let $X$ be a fair bit, encode it perfectly by $Z = X$, and choose the prior $p(Z = 0) = p(Z = 1) = 1/2$. Then the aggregated posterior already equals the prior, so $D_{\mathrm{KL}}(q(z)\|p(z)) = 0$. Yet the encoder carries one bit about $X$, and

$$I_q(X; Z) = \log 2, \qquad \mathbb{E}_X D_{\mathrm{KL}}(q(z \mid X)\|p(z)) = \log 2.$$

Matching the aggregated posterior to the prior does not force the representation to forget the input. Conversely, if the encoder ignores $x$ and always returns the prior, the KL is zero and $I_q(X; Z) = 0$. Such a representation can suffer posterior collapse if the decoder can explain the data unaided.

The rate–distortion analogy is therefore more than a slogan, but the word *rate* requires care: the KL to the prior contains mutual information *plus* aggregated-posterior mismatch. Neither term automatically measures the semantic usefulness of a latent coordinate.

## §1.11 Limits of the ELBO

The ELBO is powerful, but it is not magic. Several limitations remain.

First, the variational family may be too small. A diagonal Gaussian $q_\phi(z \mid x)$ cannot represent a complicated multimodal posterior.

Second, the prior may be too simple. The standard normal prior is convenient, but the true aggregated latent structure of data may not be standard Gaussian.

Third, the decoder may be too powerful. If the decoder can model the data without using $z$, the approximate posterior may collapse toward the prior. This is posterior collapse.

Fourth, maximizing a lower bound is not the same as exactly maximizing $\log p_\theta(x)$. The gap is the posterior KL.

## §1.12 What survives later

The VAE objective is not merely a heuristic sum of two losses. It is a variational identity:

$$\log p_\theta(x) = \mathcal{L}(\theta, \phi; x) + D_{\mathrm{KL}}(q_\phi(z \mid x)\|p_\theta(z \mid x)).$$

The ELBO can be written as

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x \mid z)] - D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z)).$$

The first form explains why it is a lower bound. The second form explains why it looks like reconstruction plus regularization.

The remaining question is computational: how can we differentiate an expectation with respect to parameters that also control the sampling distribution? That is the role of the reparameterization trick.